

Technology Services

IT Remote Access Guidelines

1. Purpose

These guidelines provide a clear direction for securely connecting to the Southern Cross University (SCU) network remotely. These guidelines aim to minimise the risk of network disruption, data loss, and unauthorised disclosure of sensitive information, including intellectual property and personally identifiable information. Adhering to these guidelines is critical to maintaining SCU's overall security posture and aligns with the Australian Cyber Security Centre Information Security Manual (ISM).

2. Scope

These guidelines apply to all members of the SCU community, including faculty, students, administrative staff, alumni, authorised guests, delegates, and independent contractors who access, implement, and manage information and information systems remotely. These guidelines do not apply to authorised and authenticated access to email, e-learning platforms, and publicly accessible university websites.

3. Risk-based approach

Remote access controls are implemented following a risk-based approach, considering the sensitivity of the information and systems being accessed. Confidentiality, Integrity, and Availability (CIA) security objectives must be applied to remote access, ensuring information is protected against unauthorised access, alteration, and destruction while providing authorised users' timely access.

4. Summary of requirements

Access Approval:	- Requests must justify the need, the duration of access, and the specific resources required. - Remote access will only be granted based on documented business needs and must be approved by the relevant Senior Manager.
Authorised Solutions:	- Remote access to SCU networks must use institutionally approved solutions, such as VPNs or browser-based SSL connections. - These solutions must use strong encryption methods, such as AES-256 for VPNs and TLS 1.2 or higher for SSL connections.
Access Limitations:	- Remote access solutions must be configured to facilitate access only to the information systems and assets the user is authorised to access. - Access rights must be based on the principle of least privilege.
Connection Duration:	- Users must disconnect immediately after completing business-related work. - Remote access sessions must automatically disconnect after inactivity (e.g., 15 minutes).
Device Security:	- Users must ensure their devices are free from malicious software and secure against unauthorised use. - Devices must have up-to-date antimalware solutions, and operating systems must be patched with the latest security updates. - Personal devices used for remote access must comply with SCU's security standards.
Multi-Factor Authentication (MFA):	All remote access must be protected by two-factor authentication using approved methods such as hardware tokens or authenticator apps.
Regular Audits and Monitoring:	- Remote access logs must be routinely monitored and inspected for compliance with security policies. - Any anomalies or suspicious activities must be reported to the IT Security team immediately for investigation.
User Training and Awareness:	- Users must complete mandatory cybersecurity training before being granted remote access. - Regular training sessions and awareness programs must be conducted to inform users about the latest security threats and best practices.

5. Implementation of Remote Access

5.1 Remote Access Solution

Remote access to SCU networks must only be undertaken through authorised solutions and systems. All university technology and security policies that apply to access from on-campus workstations also apply to workstations used for remote access. In particular, the following must be considered.

- **Authorised Solutions:** Access must be through centrally managed, institutionally approved VPNs or browser-based SSL connections. Specific encryption standards, such as AES-256 for VPNs and TLS 1.2 or higher for SSL, must be used.
- **Configuration:** Solutions should only provide access to authorised information systems and assets.
- **Approval:** No division or third party may set up remote access without documented approval for an exemption. Exemptions must be reviewed and approved by the CIO or CISO.
- **Encryption:** Data transmission must be encrypted using institutionally approved methods. All encryption keys must be managed securely, and encryption algorithms must comply with NIST standards.
- **Management:** Avoid endpoint-hosted solutions and maintain sufficient network segmentation to ensure that compromised systems do not affect other network parts.
- **Prohibited Solutions:** Do not use unauthorised remote access solutions like PCAnywhere, VNC, or TeamViewer without the CIO's or CISO's explicit approval. Such solutions, if approved, must adhere to the same security standards as authorised solutions.
- **Account Creation:** Staff will get access to the approved VPN upon account creation. Non-staff entities such as contractors must go through additional verification and authorisation processes

5.2 Additional Authorisation

Remote access for non-staff and privileged users is managed separately from staff access. The following must be considered:

- **Business Needs:** Access for non-staff and privileged users is granted based on documented business needs and must be authorised by the user's manager
- **Approval Process:** Non-staff and privileged users do not receive automatic VPN access upon account creation. They must go through an additional verification and authorisation process.
- **Security Measures:** Remote devices must have active local firewalls, up-to-date antimalware solutions, and patched operating systems. Devices must comply with SCU's Patch Management Guidelines.
- **Role Changes:** Managers must review remote access for users changing roles and revoke access if no longer required.
- **Overseas Access:** Requests must be approved before travel, and access may be revoked if security alerts arise. Specific guidelines for overseas access, including VPN usage and monitoring, must be followed.
- **Access Limitations:** Access must be restricted to specific time frames and resources necessary to perform contracted duties, with automatic expiration post-engagement. Access must adhere to the principle of least privilege.

5.3 Third Parties

Remote access by third-party users must be considered independent from remote access by staff with direct links to Southern Cross University (e.g., staff, students and faculty). It must also be possible to differentiate these users in the user management system.

- **Agreements:** Third parties must agree to SCU's access terms documented in the Third-Party Agreement, Service Partnership Agreements, and/or Service Level Agreements (SLA).
- **Access Limitations:** Restrict access to specific time frames and resources, with automatic expiration post-engagement. Access must be limited to the minimum necessary to perform contracted duties.
- **Annual Review:** Third-party access accounts must be reviewed annually to ensure active agreements and compliance with SCU's security policies.

5.4 Authentication and Access Control

SCU systems must accurately identify who is connecting remotely and verify the required access levels. This is achieved by user authentication and identification:

- **Multi-Factor Authentication (MFA):** Mandatory for all users, combining something known (password) and something possessed (token or mobile app). Acceptable methods include hardware tokens, SMS-based verification, and authenticator apps.
- **Identity Management:** Access is controlled through SCU's Identity Management System. All remote access accounts must be linked to individual identities and roles.
- **Session Management:** Inactive sessions must close after a defined period (e.g., 15 minutes of inactivity). Systems should prompt users before session termination.
- **Revocation Rights:** SCU reserves the right to revoke access if it threatens information systems or assets. Immediate revocation must occur if a security breach is detected.

6. Auditing and Monitoring

Technology Services will routinely monitor and inspect remote access logs and network traffic for evidence of compliance. Monitoring tools must include intrusion detection/prevention systems (IDS/IPS) and security information and event management (SIEM) systems.

6.1 Logging and log analysis

Effective logging and log analysis of VPN connections is vital to accounting for activities performed on a network. Effective logging also provides a central repository of information in the event of an attempted or successful compromise.

VPN connection information will be logged where available and must include:

- **Authentication Information:** Certificate details, user credentials, and remote host information.
- **Session Information:** Connection establishment time, duration, and data transferred.
- **Activities Performed:** User activities, especially those relating to sensitive resources.
- **Remote Host Information:** Operating system, IP address, MAC address, and hostname.

Analysis: Regular analysis of logs to detect and respond to suspicious activities. Retain logs for at least one year or as SCU's data retention policy requires.

7. Remote Access User Responsibility

Remote Access users must ensure that the security of their Remote Access connection is given the same consideration as their on-site connection to SCU's network. Responsibilities include:

- **Data Security:** Do not store sensitive SCU data on personal devices. Use SCU-provided secure storage solutions.
- **Incident Reporting:** Report lost devices or suspected unauthorised access immediately to Technology Services.
- **Public Wi-Fi:** Avoid using public Wi-Fi for remote access unless connected through a secure VPN.
- **Compliance:** Adhere to Patch Management Guidelines and SCU ICT Security policies and ensure personal devices meet security standards.
- **Training:** Complete mandatory cybersecurity training and awareness programs.

8. Exemptions

- **Approval:** Exemptions are granted only when compliance is not feasible, reviewed annually. Exemption requests must include justification and documentation provided to the Senior Manager of Cyber Security for review.
- **Documentation:** Justification and documentation for exemption requests must be thorough and demonstrate why compliance is impractical.

9. Roles, Responsibilities and Accountabilities

Role	Responsibility
Director, Cyber Security	• Approve any exemptions to the IT Security Policy or these Remote Access Guidelines
Cyber Resilience Team (CRT)	• Develop, maintain and improve guidelines • Monitor and reporting on compliance to measure the effectiveness. • Consult on the exemptions to the remote access policy.
Infrastructure teams	• Implement and maintain infrastructure security in line with security standards
Remote Access User	• Apply and seek approval from their manager to access VPN • Adhere to Section 6 of this document (Remote access user responsibilities)
Application Owners	• Implement and develop applications in line with security standards.
Contractors / Third Party	• Must apply for NSA (non-standard account) and use the university VPN process. • Must ensure their device complies with the security standards of the university. • Adhere to Section 6 of this document (Remote access user responsibilities)

10. Related Documents and Information

[Using Virtual Private Networks | Cyber.gov.au](#)

[Remote Access to Operational Technology Environments | Cyber.gov.au](#)

11. Document control

These guidelines shall be subject to periodic review to ensure currency and relevancy.

Document owner:		Director, Cyber Security		
Document approver:		Director, Cyber Security		
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Compliance				
Essential 8		MFA: ML1, ML2, ML3		
ISM		Principle: PROTECT P3, P4, P7, P12 Controls: ISM-0859, ISM-1321, ISM-1711, ISM-1556		
NIST		NIST 800-46		
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Appendix 1 - Definitions

Term	Definition
VPN (Virtual Private Network)	A secure network connection that allows users to access resources remotely while encrypting their data.
ISM (Information Security Manual)	The ISM is a cyber security framework produced by the Australian Signals Directorate. It outlines guidelines for organisations to protect their systems and data from cyber threats. It provides a structured approach to risk management and security practices.
NSA (Non-Standard Accounts)	Accounts created outside the regular staff or student accounts, often serving specialised roles or specific system functions.
SSO (Single Sign-On)	An authentication method enabling users to log in once and access multiple services seamlessly.
MFA (Multifactor Authentication)	MFA is a security method that requires two or more proofs of identity to grant access. It enhances security by combining different authentication factors, such as something the user knows (password), something the user has (smartphone), or something the user is (fingerprint).
IP Address (Internet Protocol Address)	An IP address is a unique numerical label assigned to a device connected to a computer network using the Internet Protocol. It serves two main functions: network interface identification and location addressing. IP addresses allow devices to communicate and exchange information on a network.
MAC address	A MAC address is a unique identifier assigned to a network interface card (NIC) or network adapter. It operates at the data link layer of the OSI model and is used for communication within a local network. Unlike IP addresses, MAC addresses are hardware-based and remain constant for a specific device.
SSL	SSL is a cryptographic protocol that ensures secure communication over the internet. It provides encryption and authentication, protecting data transmitted between a client (such as a web browser) and a server (such as a website). SSL certificates validate the authenticity of websites and enable HTTPS connections.
Antimalware	Software that detects and removes malicious programs from a computer system.
local firewall	A local firewall (also known as a personal firewall) is a Security software or hardware that filters network traffic to protect a device within a local network outgoing traffic to protect against unauthorized access and potential threats.